Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $3\sin\beta = \sin(2\alpha+\beta)$, then $\tan(\alpha+\beta) - 2\tan\alpha$ is

(A) independent of α

(B) independent of β

(C) independent of both α and β

(D) independent of none of them

Q.2

The value of $\cos 12^\circ + \cos 84^\circ + \cos 156^\circ + \cos 132^\circ$ is

(A) 1/8

(B) -1/2

(C) 1

(D) 1/2

Q.3

If A, B, C are the angles of a triangle and $\cos B + \cos C = 4\sin^2(A/2)$, then $\tan B/2 \tan C/2$ is equal to

(A) $1/2$

(B) $1/3$

(C) $2/3$

(D) $3/2$

Q.4

If $\cos\alpha + \cos\beta = a, \sin\alpha + \sin\beta = b$ and θ is the arithmetic mean between α and β then $\sin 2\theta + \cos 2\theta$ is equal to

(A) $(a+b)^2/\left(a^2+b^2\right)$

(B) $(a-b)^2/\left(a^2+b^2\right)$

(C) $\left(a^2-b^2\right)/\left(a^2+b^2\right)$

(D) none of these

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Q.5

If A, B, C are the angles of a triangle such that angle A is obtuse then

(A) $\tan A \tan B < 1$

(B) $\tan B \tan C < 1$

(C) $\tan C \tan A < 1$

(D) $\tan A \tan B \tan C < 1$

Q.6

If the angles A, B, C of a triangle are in A.P. such that $\sin(2A + B) = 1/2$ then $\sin(B + 2C) =$

(A) $-1/2$

(B) $1/2$

(C) $\sqrt{3}/2$

(D) $1/\sqrt{2}$

Q.7

Period of $\frac{\sin\theta+\sin 2\theta}{\cos\theta+\cos 2\theta}$ is

(A) 2π

(B) π

(C) $\frac{2\pi}{3}$

(D) $\frac{\pi}{3}$

Q.8

The expression $\frac{\cos 6x+6\cos 4x+15\cos 2x+10}{\cos 5x+5\cos 3x+10\cos x}$ is equal to

(A) $\cos 2x$

(B) $2\cos x$

(C) $\cos^2 x$

(D) $1 + \cos x.$

Q.9

If $f(x) = \cos(\log x)$ then
$f(x)f(y) - \frac{1}{2}\left\{f\left(\frac{x}{y}\right) + f(xy)\right\}$ is equal to :

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(A) 0

(B) 1

(C) -1

(D) none of these

Q.10

$\sin 12^\circ \sin 24^\circ \sin 48^\circ \sin 84^\circ =$

(A) $\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ$

(B) $\sin 20^\circ \sin 40^\circ \sin 60^\circ \sin 80^\circ$

(C) $\frac{3}{15}$

(D) None of these

Q.11

The expression
$\left(\frac{\cos A+\cos B}{\sin A-\sin B}\right)^n + \left(\frac{\sin A+\sin B}{\cos A-\cos B}\right)^n =$

(A) $2\cot^n\left(\frac{A-B}{2}\right)$ if n is even

(B) 0 if n is even

(C) $2\cot^n\left(\frac{A-B}{2}\right)$ if n is odd

(D) 3 if n is odd

Q.12

What is $\cos 20^\circ + \cos 100^\circ + \cos 140^\circ$ equal to ?

(A) 2

(B) 3

(C) $1/2$

(D) 0

Q.13

$\frac{\sin x-\sin 3x}{\sin^2 x-\cos^2 x}$ is equal to

(A) $\sin 2x$

(B) $\cos 2x$

(C) $\tan 2x$

(D) None of these

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Q.14

The expression $\frac{\cos 6x+6\cos 4x+15\cos 2x+10}{\cos 5x+5\cos 3x+10\cos x}$ is equal to

(A) $\cos 2x$

(B) $2\cos x$

(C) $\cos^2 x$

(D) $1+\cos x$

Q.15

In a triangle $\mathrm{ABC}, \sin \mathrm{A}-\cos \mathrm{B}=\cos \mathrm{C}$, then what is B equal to ?

(A) π

(B) $\pi/3$

(C) $\pi/2$

(D) $\pi/4$

Answers & Solutions

Q.1 Answer:

independent of both α and β

Solution:

$$\sin(2\alpha+\beta)=3\sin\beta$$
$$\Rightarrow \frac{\sin(2\alpha+\beta)+\sin\beta}{\sin(2\alpha+\beta)-\sin\beta}=\frac{3+1}{3-1}$$
$$\Rightarrow \quad \frac{2\sin(\alpha+\beta)\cos\alpha}{2\cos(\alpha+\beta)\sin\alpha}=2\Rightarrow \tan(\alpha+\beta)-2\tan\alpha=0$$

Q.2 Answer:

-1/2

Solution:

$$(\cos 12^\circ+\cos 132^\circ)+(\cos 84^\circ+\cos 156^\circ)$$
$$=2\cos\tfrac{12^\circ+132^\circ}{2}\cos\tfrac{12^\circ-132^\circ}{2}+$$
$$\quad 2\cos\tfrac{84^\circ+156^\circ}{2}\cos\tfrac{84^\circ-156^\circ}{2}$$
$$=2\cos 72^\circ\cos 60^\circ+2\cos 120^\circ\cos 36^\circ$$
$$=2\times\tfrac{\sqrt{5}-1}{4}\times\tfrac{1}{2}+2\times\left(-\tfrac{1}{2}\right)\times\tfrac{\sqrt{5}+1}{4}=-\tfrac{1}{2}.$$

Q.3 Answer:

$1/3$

Solution:

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$\cos B + \cos C = 4\sin^2(A/2)$
$\Rightarrow \quad 2\cos\frac{B+C}{2}\cos\frac{B-C}{2} = 4\sin\frac{A}{2}\sin\left(\frac{\pi}{2} - \frac{B+C}{2}\right)$
$\Rightarrow \quad 2\sin\frac{A}{2}\cos\frac{B-C}{2} = 4\sin\frac{A}{2}\cos\frac{B+C}{2}$
$\Rightarrow \cos\frac{B}{2}\cos\frac{C}{2} + \sin\frac{B}{2}\sin\frac{C}{2} = 2\left(\cos\frac{B}{2}\cos\frac{C}{2} - \sin\frac{B}{2}\sin\frac{C}{2}\right)$
$\Rightarrow \quad 3\sin\frac{B}{2}\sin\frac{C}{2} = \cos\frac{B}{2}\cos\frac{C}{2}$
$\Rightarrow \quad \tan\frac{B}{2}\tan\frac{C}{2} = \frac{1}{3}$

Q.4 Answer:

none of these

Solution:

From the given relations we have $2\cos\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2} = a$ **and** $2\sin\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2} = b$
By dividing we get $\tan\frac{\alpha+\beta}{2} = \frac{b}{a} \Rightarrow \tan\theta = \frac{b}{a}$
so that $\cos 2\theta = \frac{1-b^2/a^2}{1+b^2/a^2} = \frac{a^2-b^2}{a^2+b^2}$ **and** $\sin 2\theta = \frac{2ab}{a^2+b^2}$.
$\therefore \sin 2\theta + \cos 2\theta = \frac{a^2-b^2+2ab}{a^2+b^2}$

Q.5 Answer:

$\tan B \tan C < 1$

Solution:

$A > \pi/2 \quad B + C < \pi/2 \Rightarrow \tan(B+C) > 0$
$\Rightarrow \quad \frac{\tan B + \tan C}{1 - \tan B \tan C} > 0$
$\Rightarrow \tan B \tan C < 1 (\text{as } \tan B, \tan C > 0)$

Q.6 Answer:

$-1/2$

Solution:

$B = 60^\circ, \sin(2A+B) = 1/2 \Rightarrow 2A + B = 150^\circ$
$\Rightarrow A = 45^\circ, C = 75^\circ$
so that $B + 2C = 210^\circ = 180^\circ + 30^\circ$
$\Rightarrow \quad \sin(B+2C) = -\sin 30^\circ = -1/2.$

Q.7 Answer:

$\frac{2\pi}{3}$

Solution:

$\frac{\sin\theta + \sin 2\theta}{\cos\theta + \cos 2\theta} = \frac{2\sin\left(\frac{3\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)}{2\cos\left(\frac{3\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)} = \tan\left(\frac{3\theta}{2}\right)$
Hence period $= \frac{2\pi}{3}$

Q.8 Answer:

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$2\cos x$

Solution:

The given expression can be written as $\frac{(\cos 6x+\cos 4x)+5(\cos 4x+\cos 2x)+10(\cos 2x+1)}{\cos 5x+5\cos 3x+10\cos x}$

$= \frac{2\cos 5x\cos x+5.2\cos 3x\cos x+10.2\cos^2 x}{\cos 5x+5\cos 3x+10\cos x}$

$= \frac{2\cos x(\cos 5x+5\cos 3x+10\cos x)}{\cos 5x+5\cos 3x+10\cos x} = 2\cos x$

Q.9 Answer:

0

Solution:

Given $f(x) = \cos(\log x)$

$\therefore f(xy) = \cos(\log xy)$

$f(xy) = \cos[\log x + \log y]$

And $f\left(\frac{x}{y}\right) = \cos\left(\log \frac{x}{y}\right)$

$f\left(\frac{x}{y}\right) = \cos(\log x - \log y)$

Adding (i) and (ii), we get $f(xy) + f\left(\frac{x}{y}\right) = \cos(\log x + \log y) + \cos(\log x - \log y)$

$= 2\cos(\log x)\cdot\cos(\log y)$

$\Rightarrow f(xy) + f\left(\frac{x}{y}\right) = 2f(x).f(y)$

Then the value of $f(x)f(y) - \frac{1}{2}\left\{f\left(\frac{x}{y}\right) + f(xy)\right\}$

$= f(x)f(y) - \frac{1}{2}\cdot 2\{f(x)\cdot f(y)\} = 0$

Q.10 Answer:

$\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ$

Solution:

$\sin 12^\circ \sin 24^\circ \sin 48^\circ \sin 84^\circ$

$= \frac{1}{4}(2\sin 12^\circ \sin 48^\circ)(2\sin 24^\circ \sin 84^\circ)$

$= \frac{1}{2}(\cos 36^\circ - \cos 60^\circ)(\cos 60^\circ - \cos 108^\circ)$

$= \frac{1}{4}\left(\cos 36^\circ - \frac{1}{2}\right)\left(\frac{1}{2} + \sin 18^\circ\right)$

$= \frac{1}{4}\left\{\frac{1}{4}(\sqrt{5}+1) - \frac{1}{2}\right\}\left\{\frac{1}{2} + \frac{1}{4}(\sqrt{5}-1)\right\} = \frac{1}{16}$

and $\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ$

$= \frac{1}{2}[\cos(60^\circ - 20^\circ)\cos 20^\circ \cos(60^\circ + 20^\circ)]$

$= \frac{1}{2}\left[\frac{1}{4}\cos 3(20^\circ)\right] = \frac{1}{8}\cos 60^\circ = \frac{1}{2}\times\frac{1}{8} = \frac{1}{16}.$

Q.11 Answer:

$2\cot^n\left(\frac{A-B}{2}\right)$ if n is even

Solution:

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The given expression

$$= \left(\frac{2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)}{2\cos\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)}\right)^n + \left(\frac{2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)}{2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{B-A}{2}\right)}\right)^n$$

$$= \cot^n\left(\frac{A-B}{2}\right) + (-1)^n \cot^n\left(\frac{A-B}{2}\right)$$

$$= \begin{cases} 2\cot^n\left(\frac{A-B}{2}\right), & \text{if n is even} \\ 0 & \text{, if n is odd} \end{cases}$$

Q.12 Answer:

0

Solution:

$$\cos 20^\circ + \cos 100^\circ + \cos 140^\circ$$
$$= (\cos 140^\circ + \cos 20^\circ) + \cos 100^\circ$$
$$= 2\cos\left(\frac{160^\circ}{2}\right)\cdot\cos\left(\frac{120^\circ}{2}\right) + \cos 100^\circ$$
$$= 2\cdot\cos 80^\circ\cdot\frac{1}{2} + \cos 100^\circ$$
$$= 2\cos\left(\frac{180^\circ}{2}\right)\cdot\cos\left(\frac{20^\circ}{2}\right)$$
$$= 2\cos 90^\circ\cdot\cos 10^\circ$$
$$= 2\times 0\times\cos 10^\circ = 0$$

Q.13 Answer:

None of these

Solution:

$$\frac{\sin x-\sin 3x}{\sin^2 x-\cos^2 x} = \frac{\sin 3x-\sin x}{\cos^2 x-\sin^2 x}$$
$$= \frac{2\cos\frac{3x+x}{2}\cdot\sin\frac{3x-x}{2}}{\cos 2x} = \frac{2\cos 2x\cdot\sin x}{\cos 2x}$$
$$= 2\sin x$$

Q.14 Answer:

$2\cos x$

Solution:

The given expression can be written as $(\cos 6x+\cos 4x)+5(\cos 4x+\cos 2x)+10(\cos 2x+1)$

$$\cos 5x+5\cos 3x+10\cos x$$
$$= \frac{2\cos 5x\cos x+5.2\cos 3x\cos x+10.2\cos^2 x}{\cos 5x+5\cos 3x+10\cos x}$$
$$= \frac{2\cos x(\cos 5x+5\cos 3x+10\cos x)}{\cos 5x+5\cos 3x+10\cos x} = 2\cos x$$

Q.15 Answer:

$\pi/2$

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Solution:

In a $\Delta \mathrm{ABC}$**, we have** $\sin A-\cos B=\cos C \Rightarrow \sin A=\cos B+\cos C$

$\Rightarrow 2 \sin \frac{\mathrm{A}}{2} \cdot \cos \frac{\mathrm{A}}{2}$

$=2 \cos \left(\frac{\mathrm{B}+\mathrm{C}}{2}\right) \cdot \cos \left(\frac{\mathrm{B}-\mathrm{C}}{2}\right)$

$\left[\because \sin 2 \mathrm{~A}=2 \sin \mathrm{A} \cdot \cos \mathrm{A}\right]$

and $\cos B+\cos C=2 \cos \left(\frac{B+C}{2}\right) \cdot \cos \left(\frac{B-C}{2}\right)$

$\Rightarrow 2 \sin \frac{\mathrm{A}}{2} \cdot \cos \frac{\mathrm{A}}{2}=2 \cos \left(90^{\circ}-\frac{\mathrm{A}}{2}\right) \cdot \cos \left(\frac{\mathrm{B}-\mathrm{C}}{2}\right)$

$\left[\because \mathrm{A}+\mathrm{B}+\mathrm{C}=180^{\circ} \Rightarrow\left(\frac{\mathrm{B}+\mathrm{C}}{2}\right)=90^{\circ}-\frac{\mathrm{A}}{2}\right]$

$\Rightarrow 2 \sin \frac{\mathrm{A}}{2} \cdot \cos \frac{\mathrm{A}}{2}=2 \sin \frac{\mathrm{A}}{2} \cdot \cos \left(\frac{\mathrm{B}-\mathrm{C}}{2}\right)$

$\left[\because \cos \left(90^{\circ}-\theta\right)=\sin \theta\right]$

$\Rightarrow \cos \frac{\mathrm{A}}{2}=\cos \left(\frac{\mathrm{B}-\mathrm{C}}{2}\right)$

$\Rightarrow \frac{\mathrm{A}}{2}=\frac{\mathrm{B}-\mathrm{C}}{2}$

$\Rightarrow \mathrm{A}+\mathrm{C}=\mathrm{B}$

Also, $\mathrm{A}+\mathrm{C}=180^{\circ}-\mathrm{B}$

So, $180^{\circ}-\mathrm{B}=\mathrm{B}$

$\Rightarrow 2 \mathrm{~B}=180^{\circ} \quad \therefore \mathrm{B}=90^{\circ}$

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